

驾驶意图增强面向规划的强化学习

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摘要

在每个场景仅使用单条示教轨迹训练的连续动作策略会遭受模式坍缩：采样结果聚集在示教的操纵动作周围，导致策略无法表示语义上不同的替代方案。在基于偏好的评估下，这会限制最佳- N 性能的上限——即使先知选择也无法恢复采样分布中不包含的内容。我们提出DIAL，一个两阶段驾驶意图增强强化学习框架，用于偏好对齐的连续动作驾驶策略。在第一阶段，DIAL通过无分类器引导将流匹配动作头条件化为离散意图标签，从而扩展了采样分布以覆盖不同的操控模式，并打破了单演示模式坍缩。在第二阶段，DIAL通过多意图群体相对策略优化将此扩展后的分布引入偏好强化学习，该优化在每个偏好组内覆盖所有意图类别，并防止微调过程重新坍缩到当前偏好的模式。以八种规则推导的意图实例化为端到端驾驶，并在WOD-E2E上评估：具有竞争力的视觉到动作和视觉-语言-动作监督微调基线在最佳-128时均低于人类驾驶演示的性能上限，最强的先前方法在最佳-64时仅达到RFS 8.5；意图CFG采样将这一上限提升至最佳-128时的RFS 9.14，首次同时超越了先前最佳方法和人类驾驶演示；多意图GRPO将留出RFS从7.681提升至8.211，而每个单意图基线的峰值均更低且到训练结束时发生退化。这些结果表明，基于演示数据训练的连续动作策略上进行偏好强化学习的瓶颈不仅在于如何更新策略，更在于如何扩展并保持被优化的采样分布。

1 引言

演示数据记录的是实际发生的情况，而非本应发生的情况。一条记录轨迹是在特定场景中实现的一种物理上可行的未来。在决策做出之前，可能仍有多种未来是可采纳的：车辆可以更早或更晚刹车、保持车道或准备并道、果断前行或保守让行。演示数据认证的是已执行的一种行为，而非该行为是唯一安全的或人类评估者最为偏好的。当连续动作策略被训练为几何模仿学习时，这种区别很容易被忽略，但一旦策略通过偏好对齐而非与单一记录日志的距离来评估时，它就变得至关重要。

在机器人操作、导航和端到端驾驶领域 [Zitkovich 等人, 2023, Kim 等人, 2024, Zhou 等人, 2025, Black 等人, 2024, Chai 等人, 2025], 连续动作策略通常通过对每个场景的单条示教轨迹进行模仿学习或监督微调来训练。借助具有表现力的流匹配或扩散动作头，这种方法能在示教轨迹附近产生平滑的局部控制，

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Driving Intents Amplify Planning-Oriented Reinforcement Learning

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Abstract

Continuous-action policies trained on a single demonstrated trajectory per scene suffer from *mode collapse*: samples cluster around the demonstrated maneuver and the policy cannot represent semantically distinct alternatives. Under preference-based evaluation, this caps best-of- N performance – even oracle selection cannot recover what the sampling distribution does not contain. We introduce DIAL, a two-stage **Driving-Intent-Amplified reinforcement Learning** framework for preference-aligned continuous-action driving policies. In the first stage, DIAL conditions the flow-matching action head on a discrete intent label with classifier-free guidance (CFG), which expands the sampling distribution along distinct maneuver modes and breaks single-demonstration mode collapse. In the second stage, DIAL carries this expanded distribution into preference RL through *multi-intent Group Relative Policy Optimization (GRPO)*, which spans all intent classes within every preference group and prevents fine-tuning from re-collapsing around the currently preferred mode. Instantiated for end-to-end driving with eight rule-derived intents and evaluated on WOD-E2E: competitive Vision-to-Action (VA) and Vision-Language-Action (VLA) Supervised Finetuning (SFT) baselines plateau below the human-driven demonstration at best-of-128, with the strongest prior (RAP) capping at ater Feedback Score (RFS) 8.5 even with best-of-64; intent-CFG sampling lifts this ceiling to RFS 9.14 at best-of-128, surpassing both the prior best (RAP, 8.5) and the human-driven demonstration (8.13) for the first time; and multi-intent GRPO improves held-out RFS from 7.681 to 8.211, while every single-intent baseline peaks lower and degrades by training end. These results suggest that the bottleneck of preference RL on continuous-action policies trained from demonstrations is not only how to update the policy, but how to expand and preserve the sampling distribution being optimized.

1 Introduction

Demonstrations record what happened, not what should have happened. A logged trajectory is one physically feasible future realized in a particular scene. Before the decision was made, several futures may still have been admissible: the vehicle may brake earlier or later, keep its lane or prepare a merge, proceed assertively or yield conservatively. The demonstration certifies one executed behavior, not that this behavior is uniquely safe or most preferred by human evaluators. That distinction is easy to miss when continuous-action policies are trained as geometric imitation, but it becomes central once policies are evaluated by preference alignment rather than by distance to a single recorded log.

Across robotic manipulation, navigation, and end-to-end driving [Zitkovich et al., 2023, Kim et al., 2024, Zhou et al., 2025, Black et al., 2024, Chi et al., 2025], continuous-action policies are typically trained by imitation or supervised fine-tuning on a single demonstrated trajectory per scene. With expressive flow-matching or diffusion action heads, this produces smooth local control near the

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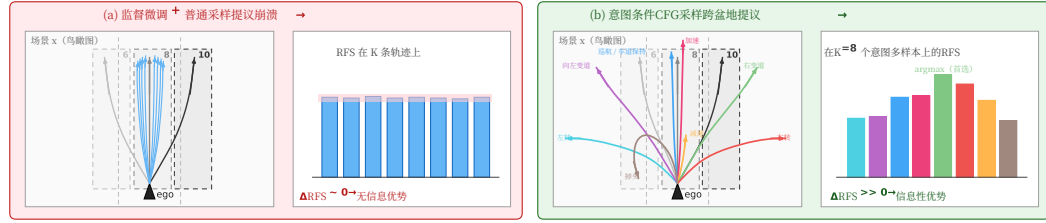


图1: 驾驶意图通过暴露场景内偏好对比来增强面向规划的强化学习。(a)在SFT的 +普通采样下, K rollout坍缩至一个操作盆地, 其RFS分数几乎相同 ($\Delta RFS \approx 0$), 因此组相对优势缺乏信息量。(b)在意图条件CFG采样下, $K=8$ rollout (每个驾驶意图一个) 分散至不同的操作盆地, 其RFS分数分布广泛 ($\Delta RFS \gg 0$), 从而暴露出面向规划的强化学习所放大的偏好对比。

但对于暴露替代操作几乎没有作用: 采样轨迹会聚集在记录路径周围, 导致策略坍缩为单一行为模式。多模态轨迹表示长期以来一直用于解决预测问题 [Chai 等人, 2019, Phan-Minh 等人, 2020, Salzmänn 等人, 2020, Shi 等人, 2022], 但对于动作规划器的自车策略而言, 单一示范监督仍属默认做法。当策略通过到示范的距离进行评分时, 模式坍缩很难被发现, 因为该指标恰恰奖励了坍缩所集中的行为。基于偏好的评估使其可视化。WOD-E2E RFS 基准 [Xu 等, 2025a] 根据多个人工标注的替代方案对预测轨迹进行评分, 因此记录的真实值是一个被评分的候选方案而非最优目标, 而策略样本上 best-of- K RFS 则量化了在先知选择下可达到的评分者偏好质量。在四个具有竞争力的监督微调视觉-动作和监督微调视觉语言动作基线中 (图3), best-of- K RFS 在 $K=128$ 处饱和, 低于记录的真实值 (RFS 8.13): 一个模式坍塌策略无论样本预算如何都无法达到评分者偏好区域。

因此, 我们提出 DIAL, 即驾驶意图增强的强化学习, 这是一个两阶段训练框架, 首先扩展连续动作驾驶策略的采样分布, 然后在偏好强化学习过程中保持这种扩展。第一阶段使用意图条件CFG将同一场景解码为语义上不同的操作模式——转弯与让行、变道与保持车道、加速与制动——从而使流匹配动作头能够产生超越单条记录路径坐标级扰动的替代方案。借助八种规则推导的意图 (巡航、左/右变道、左/右转、掉头、加速、减速), 意图条件 best-of- K 在 $K \approx 8$ 处已经匹配了人类驾驶演示, 并且汇集所有八种意图的提议在 $K=128$ 处达到了 RFS 9.14。此前最强的端到端驾驶规划器 RAP [Feng 等, 2025], 即使在基于学习到的评分器进行 best-of-64 的情况下, 也仅能达到 RFS 8.5 的上限; 意图条件采样首次同时超越了这一先前上限和人类驾驶演示 (RFS 8.13)。

DIAL的第二阶段是必要的, 因为扩展的最佳- N 上限并不能直接产生可部署的策略: 推理时无法进行先知选择, 且策略仍然每帧只生成一条轨迹。强化微调是捕获该上限的自然工具, 但标准的GRPO无法保持这种扩展性。从单一意图 (无论是真实值、预测值还是最高评分) 中采样的rollout会围绕一个模态重新坍缩rollout组, 使得组相对优势缺乏偏好对比 [Schulman等人, 2017, Shao等人, 2024b]。瓶颈在于采样分布, 而非策略更新, 如图1所示。

为了保持第一阶段产生的多样性, DIAL采用多意图GRPO。在第一阶段, 我们将流匹配动作头条件设定为八种驾驶意图, 并使用无分类器引导丢弃进行训练。在第二阶段, 对每个场景, 我们在每种意图下采样少量轨迹, 将八个意图条件rollout集汇集成一个组, 并在汇集集上计算组相对优势。在部署时, 学习到的意图分类器选择意图, 并由同一个意图条件生成器解码最终轨迹, 因此RL期间使用的提议机制在推理时仍然有效。

在RFS标注的WOD-E2E验证池的确定性338/100划分上, DIAL将留出RFS从7.681提升至8.211 (相比其监督微调初始化), 而每一个单意图变体则先是达到峰值,

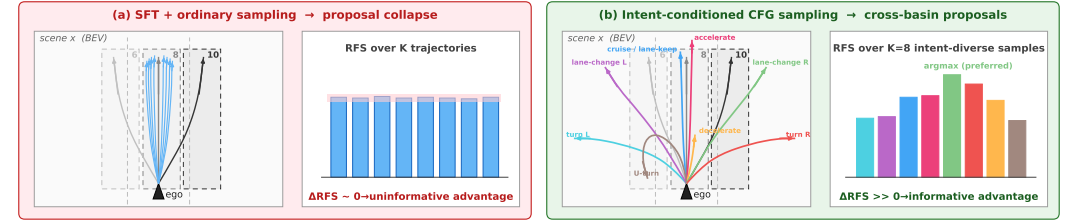


Figure 1: **Driving intents amplify planning-oriented RL by exposing within-scene preference contrast.** (a) Under SFT + ordinary sampling, K rollouts collapse into one maneuver basin and their RFS scores are nearly identical ($\Delta RFS \approx 0$), so the group-relative advantage is uninformative. (b) Under intent-conditioned CFG sampling, $K=8$ rollouts (one per driving intent) spread across distinct basins and their RFS scores spread widely ($\Delta RFS \gg 0$), exposing the preference contrast that planning-oriented RL amplifies.

demonstration but does little to expose alternative maneuvers: sampled trajectories cluster around the logged path, and the policy collapses to a single behavioral mode. Multimodal trajectory representations have long addressed this for prediction [Chai et al., 2019, Phan-Minh et al., 2020, Salzmänn et al., 2020, Shi et al., 2022], but for the ego policy of an action planner, single-demonstration supervision remains the default.

Mode collapse is hard to see when policies are scored by distance to the demonstration, because the metric rewards exactly the behavior collapse concentrates on. Preference-based evaluation makes it visible. The WOD-E2E RFS benchmark [Xu et al., 2025a] scores predicted trajectories against multiple human-annotated alternatives, so the logged GT is one scored candidate rather than an oracle target, and best-of- K RFS over a policy’s samples quantifies the rater-preferred quality reachable under oracle selection. Across four competitive SFT VA and VLA baselines (Figure 3), best-of- K RFS saturates *below* the logged G round Truth (GT) (RFS 8.13) even at $K=128$: a mode-collapsed policy does not reach the rater-preferred regions, regardless of sample budget.

Therefore, We propose DIAL, Driving-Intent-Amplified reinforcement Learning, a two-stage training framework that first expands the sampling distribution of a continuous-action driving policy and then preserves this expansion during preference RL. The first stage uses intent-conditioned CFG to decode the same scene into semantically distinct maneuver modes – turning versus yielding, changing lanes versus keeping the lane, accelerating versus braking – so the flow-matching action head produces alternatives beyond coordinate-level perturbations of one logged path. With eight rule-derived intents (cruise, lane change L/R, turn L/R, U-turn, accelerate, decelerate), intent-conditioned best-of- K already matches the human-driven demonstration at $K \approx 8$, and pooling proposals across all eight intents reaches RFS 9.14 at $K=128$. The strongest prior end-to-end driving planner, RAP [Feng et al., 2025], caps at RFS 8.5 even with best-of-64 over a learned rater; intent-conditioned sampling surpasses for the first time both this prior ceiling and the human-driven demonstration (RFS 8.13).

The second stage of DIAL is needed because an expanded best-of- N ceiling does not directly produce a deployable policy: oracle selection is unavailable at inference, and the policy still commits to one trajectory per frame. Reinforcement fine-tuning is the natural tool for capturing the ceiling, but standard GRPO does not preserve the expansion. Rollouts drawn from a single intent – whether the GT, predicted, or top-rated – re-collapse the rollout group around one mode, leaving group-relative advantages without preference contrast [Schulman et al., 2017, Shao et al., 2024b]. The bottleneck is the sampling distribution, not the policy update, as shown in Figure 1.

To preserve the diversity created by the first stage, DIAL uses *multi-intent GRPO*. In the first stage, we condition the flow-matching action head on the eight driving intents and train with classifier-free guidance dropout. In the second stage, for each scene we sample a small number of trajectories under each intent, pool the eight intent-conditioned rollout sets into a single group, and compute group-relative advantages over the pooled set. At deployment, a learned intent classifier selects the intent and the same intent-conditioned generator decodes the final trajectory, so the proposal mechanism used during RL remains active at inference.

On a deterministic 338/100 split of the RFS-labeled WOD-E2E validation pool, DIAL improves held-out RFS from 7.681 to 8.211 over its SFT initialization, while every single-intent variant peaks

随后在最终检查点出现下降。对于从演示数据训练的连续动作策略进行偏好对齐，其问题并不在于更准确地模仿记录的轨迹，也不在于使用更强的优化器；而在于摆脱模式坍塌，并通过偏好强化学习维持这一脱离状态。

贡献。

1. 我们识别出单演示监督微调模式崩塌是连续动作策略上偏好强化学习的关键瓶颈。在 WOD-E2E 上，即使进行 $K=128$ ，竞争性 SFT 基线的 best-of- N RFS 上限仍低于记录的人类驾驶演示，表明偏好优化受限于策略能采样的范围。2. 我们证明，意图条件无分类器引导通过沿语义上不同的操作模式扩展流匹配动作头，打破了这一采样上限。借助八种规则推导的驾驶意图，意图 CFG 采样达到了 9.14 的最佳-128 RFS，首次超越了先前最优规划器 RAP（RFS 8.5，最佳-64）和人类驾驶演示（RFS 8.13）。3. 我们引入了 DIAL——驾驶意图增强学习，这是一个两阶段训练框架，将意图-CFG 提案扩展与多意图 GRPO 多样性保持相结合。通过在每个偏好组内覆盖所有意图类别，DIAL 在强化学习期间保持了扩展采样分布，并将留出 RFS 从 7.681 提升至 8.211，而每个单意图变体的峰值均较低，且在训练结束时出现下降。

2 相关工作

DIAL 连接了四条工作路线。端到端自动驾驶已从基于传感器融合的模仿学习和可扩展解码，转向面向规划、向量化、稀疏化、生成式和基于扩散的规划器 [Hu 等人, 2023, Jiang 等人, 2023, Chen 等人, 2024, Sun 等人, 2025, Zheng 等人, 2024, Liao 等人, 2025, Zou 等人, 2025, Hu 等人, 2022, Chitta 等人, 2022, Wu 等人, 2022, Shao 等人, 2023, Jia 等人, 2023]。这些系统将规划作为核心输出，而 WOD-E2E 则将评估转向评分者偏好分数，而非仅依赖记录轨迹模仿 [Xu 等, 2025a]。

视觉-语言-动作模型将感知、语言条件表示和连续控制结合起来，用于机器人技术和驾驶领域 [Zitkovich 等人, 2023, Kim 等人, 2024, Black 等人, 2024, Zhou 等人, 2025]。在自动驾驶中，相关的 VLM、大语言模型和 VLA 系统研究了可解释驾驶、图 VQA、闭环语言条件驾驶、多模态驾驶语言模型、知识驱动驾驶、反事实推理数据集、推理到规划的强化学习、潜在动作预测、场景自适应专家、风格感知动作、世界模型和 VLA 推理 [Xu 等人, 2024, Sima 等人, 2024, Shao 等人, 2024a, Ma 等人, 2024, Mao 等人, 2023, Wen 等人, 2023, Wang 等人, 2025, Zhou 等人, 2026, Li 等人, 2026, Zhang 等人, 2025b, Rawal 等人, 2026, Xie 等人, 2026, Luo 等人, 2026, You 等人, 2026, Gao 等人, 2026, Zhang 等人, 2025a, Wang 等人, 2026, Li 等人, 2025a, Ye 等人, 2025]。我们的工作并非主要构建一个更大的驾驶 VLA；它着眼于在何种情况下偏好优化能够拥有可供排序的有机动级方案。

多模态驾驶预测长期以来通过锚点、轨迹集、潜在模式或意图查询 [Chai 等人, 2019, Phan-Minh 等人, 2020, Salzmann 等人, 2020, Shi 等人, 2022] 来表示未来行为。我们将意图用于不同的角色，即扩大连续自行车策略可用的机动级提议支撑集。

该优化建立在 PPO/RLHF 风格的偏好学习和组相对策略更新之上 [Schulman 等人, 2017, Ziegler 等人, 2019, Rafailov 等人, 2023, Shao 等人, 2024b]，以及近期的扩散和流策略优化 [Chi 等人, 2025, Lipman 等人, 2022, Ren 等人, 2024, Alles 等人, 2025, Zhang 等人, 2025c, McAllister 等人, 2025, Black 等人, 2023, Yang 等人, 2024, Wang 等人, 2022, Huang 等人, 2026a]。我们关注的重点是面向规划的连续动作 VLA 驾驶中，意图多样性提议与 RFS 引导的强化学习之间的交互作用。更全面的讨论见附录 A。

lower and declines by its final checkpoint. Preference alignment for continuous-action policies trained from demonstrations is not the problem of imitating logged trajectories more accurately, nor of running a stronger optimizer; it is the problem of escaping mode collapse and preserving the escape through preference RL.

Contributions.

1. We identify single-demonstration SFT mode collapse as a key bottleneck for preference RL on continuous-action policies. On WOD-E2E, the best-of- N RFS ceiling of competitive SFT baselines saturates below the logged human-driven demonstration even at $K=128$, showing that preference optimization is limited by what the policy can sample.
2. We show that intent-conditioned classifier-free guidance breaks this sampling ceiling by expanding the flow-matching action head along semantically distinct maneuver modes. With eight rule-derived driving intents, intent-CFG sampling reaches best-of-128 RFS of 9.14, surpassing for the first time both the prior best planner RAP (RFS 8.5 with best-of-64) and the human-driven demonstration (RFS 8.13).
3. We introduce DIAL, Driving-Intent-Amplified Learning, a two-stage training framework that combines intent-CFG proposal expansion with multi-intent GRPO diversity preservation. By spanning all intent classes within every preference group, DIAL preserves the expanded sampling distribution during RL and improves held-out RFS from 7.681 to 8.211, while every single-intent variant peaks lower and declines by the end of training.

2 Related Work

DIAL connects four lines of work. End-to-end autonomous driving has moved from sensor-fusion imitation and scalable decoding toward planning-oriented, vectorized, sparse, generative, and diffusion-based planners [Hu et al., 2023, Jiang et al., 2023, Chen et al., 2024, Sun et al., 2025, Zheng et al., 2024, Liao et al., 2025, Zou et al., 2025, Hu et al., 2022, Chitta et al., 2022, Wu et al., 2022, Shao et al., 2023, Jia et al., 2023]. These systems motivate planning as the central output, while WOD-E2E turns evaluation toward rater-preference scores rather than logged-trajectory imitation alone [Xu et al., 2025a].

Vision-language-action models couple perception, language-conditioned representations, and continuous control for robotics and driving [Zitkovich et al., 2023, Kim et al., 2024, Black et al., 2024, Zhou et al., 2025]. In autonomous driving, related VLM, LLM, and VLA systems study interpretable driving, graph VQA, closed-loop language-conditioned driving, multimodal driving language models, knowledge-driven driving, counterfactual reasoning datasets, reasoning-to-planning RL, latent-action prediction, scene-adaptive experts, style-aware actions, world models, and VLA reasoning [Xu et al., 2024, Sima et al., 2024, Shao et al., 2024a, Ma et al., 2024, Mao et al., 2023, Wen et al., 2023, Wang et al., 2025, Zhou et al., 2026, Li et al., 2026, Zhang et al., 2025b, Rawal et al., 2026, Xie et al., 2026, Luo et al., 2026, You et al., 2026, Gao et al., 2026, Zhang et al., 2025a, Wang et al., 2026, Li et al., 2025a, Ye et al., 2025]. Our work is not primarily a larger driving VLA; it isolates when preference optimization has useful maneuver-level proposals to rank.

Multimodal driving prediction has long represented future behavior through anchors, trajectory sets, latent modes, or intention queries [Chai et al., 2019, Phan-Minh et al., 2020, Salzmann et al., 2020, Shi et al., 2022]. We use intent for a different role, namely to enlarge the maneuver-level proposal support available to a continuous ego policy.

The optimization builds on PPO/RLHF-style preference learning and group-relative policy updates [Schulman et al., 2017, Ziegler et al., 2019, Rafailov et al., 2023, Shao et al., 2024b], as well as recent diffusion and flow policy optimization [Chi et al., 2025, Lipman et al., 2022, Ren et al., 2024, Alles et al., 2025, Zhang et al., 2025c, McAllister et al., 2025, Black et al., 2023, Yang et al., 2024, Wang et al., 2022, Huang et al., 2026a]. Our focus is the interaction between intent-diverse proposals and RFS-guided reinforcement learning for planning-oriented continuous-action VLA driving. A fuller discussion is provided in Appendix A.

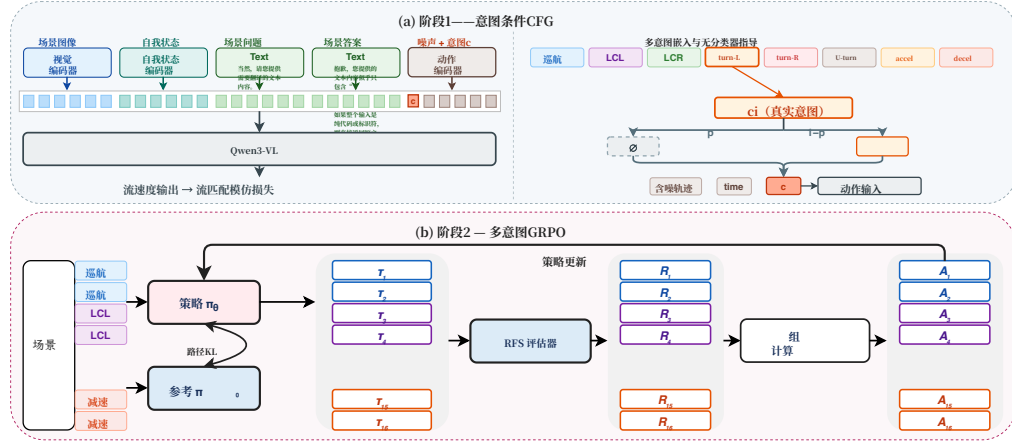


图2: DIAL 概述。 (a) 阶段1 — CFG模仿训练。扩散动作头以离散意图 c_i 为条件；CFGdropout (p_{drop}) 使模型同时学习有条件和无条件的动作分布。 (b) 阶段2 — 多意图 GRPO。每个场景中， $K=16$ 条轨迹被采样为 $S=2$ 个 rollout $\times |\mathcal{C}|=8$ 个意图，由 RFS 评估器评分，并用于通过 GRPO 相对 SFT参考 π_0 进行策略更新。

3 方法

DIAL是一个建立在MINDVLA-U1 [Huang 等人, 2026b]之上的两阶段训练流程。给定一个驾驶场景 x ，策略生成一条未来轨迹 $\tau \in \mathbb{R}^{T \times d}$ ，在RFS监督下，记录轨迹是一个评分候选项，而非最优目标。阶段1通过无分类器指导将扩散动作头以离散驾驶意图为条件，沿意图级维度扩展采样分布。阶段2在意图平衡的 rollout 组上运行GRPO，该组覆盖每个场景的完整意图集，通过偏好优化保持意图多样性。在推理时，一个轻量级意图分类器从视觉上下文中预测意图，并由同一意图条件生成器解码出最终轨迹。

3.1 意图条件CFG

我们将流匹配动作头以离散的驾驶意图 c 作为条件。意图集特意设置得很小：巡航、向左/向右变道、左转/右转、掉头、加速和减速。在模仿训练期间，这些标签通过位移、航向变化、横向偏移和速度变化，根据轨迹几何来推断。在语言模型预测流速度之前，一个意图嵌入会被添加到带噪声的动作/时间后缀中。无分类器指导的随机失活也会用无条件占位符替换掉部分意图标签，这样同一个生成器既能学习条件动作分布，也能学习无条件动作分布。在推理时，一个轻量级的前缀分类器根据视觉状态上下文预测意图，而CFG则将条件速度预测与无条件速度预测结合起来。

这一阶段的目的并非编码一套完整的驾驶本体。其目的是防止单轨迹的监督微调将不同的意图盆地坍塌成同一种延续。面对同一场景，改变 c 应当在评估者能判断的层面改变提议，而不仅仅是增加坐标噪声。

3.2 多意图 GRPO

对于每个场景，偏好优化应用于跨越整个意图集的意图平衡的提议组，而不是从单个意图中抽取的多个噪声种子。我们为每个意图采样 S 条轨迹，从而得到

$$K = |\mathcal{C}|S \quad (1)$$

建议。当前实现使用 $|\mathcal{C}| = 8$ 和 $S = 2$ ，因此每个场景贡献 $K = 16$ 条轨迹。每条轨迹由流匹配头的随机SDE采样器生成，其累积路径对数概率被存储以用于策略比率重放。

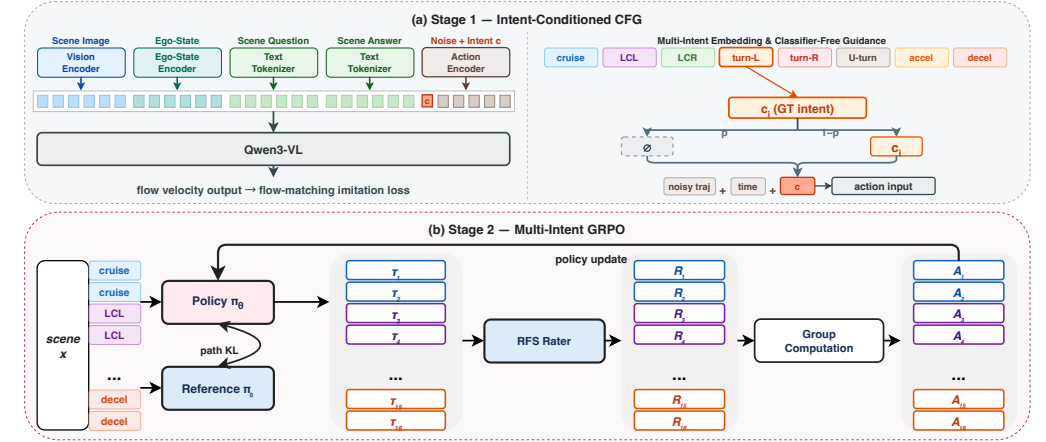


Figure 2: **Overview of DIAL.** (a) **Stage 1 — CFG Imitation Training.** The diffusion action head is conditioned on a discrete intent c_i ; CFG dropout (p_{drop}) teaches the model both conditional and unconditional action distributions. (b) **Stage 2 — Multi-Intent GRPO.** Per scene, $K=16$ trajectories are sampled as $S=2$ rollouts $\times |\mathcal{C}|=8$ intents, scored by the RFS rater, and used to update the policy via GRPO against the SFT reference π_0 .

3 Method

DIAL is a two-stage training pipeline built on top of MINDVLA-U1 [Huang et al., 2026b]. Given a driving scene x , the policy produces a future trajectory $\tau \in \mathbb{R}^{T \times d}$, and under RFS supervision the logged trajectory is one scored candidate rather than an oracle target. Stage 1 conditions the diffusion action head on a discrete driving intent with classifier-free guidance, expanding the sampling distribution along intent-level axes. Stage 2 runs GRPO over an intent-balanced rollout group that spans the full intent set per scene, preserving intent diversity through preference optimization. At inference, a lightweight intent classifier predicts the intent from visual context, and the same intent-conditioned generator decodes the final trajectory.

3.1 Intent-Conditioned CFG

We condition the flow-matching action head on a discrete driving intent c . The intent set is deliberately small: cruise, lane change left/right, turn left/right, U-turn, accelerate, and decelerate. During imitation training, these labels are inferred from trajectory geometry using displacement, heading change, lateral shift, and speed change. An intent embedding is added to the noisy-action/time suffix before the language model predicts the flow velocity. Classifier-free dropout also replaces some intent labels with an unconditional placeholder, so the same generator learns both conditional and unconditional action distributions. At inference, a lightweight prefix classifier predicts the intent from the visual-state context and CFG combines conditional and unconditional velocity predictions.

This stage is not meant to encode a complete driving ontology. Its purpose is to prevent single-trajectory SFT from collapsing distinct intent basins into one continuation. For the same scene, changing c should change the proposal at the level raters can judge, not merely add coordinate noise.

3.2 Multi-Intent GRPO

For each scene, preference optimization is applied to an intent-balanced proposal group that spans the entire intent set, not to multiple noise seeds drawn from a single intent. We sample S trajectories per intent, giving

$$K = |\mathcal{C}|S \quad (1)$$

proposals. The current implementation uses $|\mathcal{C}| = 8$ and $S = 2$, so each scene contributes $K = 16$ trajectories. Each trajectory is generated by the stochastic SDE sampler of the flow-matching head, and its cumulative path log-probability is stored for policy-ratio replay.

奖励即为评分员反馈分数：

$$R_i = \text{RFS}(\tau_i; \mathcal{P}_x), \quad (2)$$

其中 $\mathcal{P}_x$ 表示场景 x 的评分者轨迹和标签。可以包含辅助几何代价，但主要实现将其保持为零，并使用下文描述的偏好奖励。优势度在同一场景的提议内进行归一化：

$$A_i = \frac{R_i - \frac{1}{K} \sum_{j=1}^K R_j}{\text{std}_{j=1}^K(R_j) + \epsilon}. \quad (3)$$

这一步使得意图多样性对强化学习训练变得有用：如果所有提议都表达相同的意图，则该组几乎没有偏好对比；如果意图展现出不同意图，RFS会提供有意义的场景内排序。由于该组始终涵盖所有八个意图类别而非 K 同一意图的噪声变体，因此每个意图类别都会在同一次更新中收到偏好信号。

3.3 奖励篡改感知RFS奖励

报告的评估指标仍然是标准WOD-E2E RFS分数。然而，在训练中直接使用未经修改的评估器会产生两条可避免的奖励破解路径。标准RFS仅对3秒和5秒锚点进行评分，因此中间航点可能会发生漂移，而两个锚点仍保持在可接受的方框内。此外，它对最多三个评分者轨迹应用了硬最大值，因此模型可能会在强化学习训练划分上过拟合到最容易获得高标签的评分员几何结构。

因此，我们采用一种训练端RFS变体，使用相同的评分员偏好输入，但采用不同的聚合方式。首先，评分员最大值被替换为标签软最大值聚合。对于评分员标签 y_p 、几何衰减 $d_{p,a}$ 、锚点 a 和温度 τ ，每锚点分数为：

$$\tilde{R}_a = \sum_p \frac{\exp(\tau y_p)}{\sum_q \exp(\tau y_q)} y_p d_{p,a}. \quad (4)$$

权重仅取决于固定的评分员标签，而非模型控制的几何特性，因此策略无法通过向某一位评分员靠拢来操纵聚合。我们根据划分感知验证协议调整 τ ，并在第4.6.2节报告其敏感性。其次，训练锚点从 $\{3, 5\}$ 秒加密至 $\{1, 2, 3, 4, 5\}$ 秒。额外的锚点会对不连续或不合理的中间运动施加惩罚，同时保留评估所使用的标准3秒和5秒评分点。

3.4 策略更新

每条采样轨迹在生成时所使用的同一意图条件下进行回放。设 $\log p_{\theta_{\text{old}}}(\tau_i \mid x, c_i)$ 为存储的SDE路径对数概率， $\log p_{\theta}(\tau_i \mid x, c_i)$ 为回放后的当前对数概率。裁剪后的GRPO目标为：

$$\rho_i = \exp(\log p_{\theta}(\tau_i \mid x, c_i) - \log p_{\theta_{\text{old}}}(\tau_i \mid x, c_i)), \quad (5)$$

$$\mathcal{L}_{\text{GRPO}} = -\frac{1}{K} \sum_{i=1}^K \min(\rho_i A_i, \text{clip}(\rho_i, 1 - \epsilon_{\ell}, 1 + \epsilon_h) A_i). \quad (6)$$

我们使用对称裁剪，参数为 $\epsilon_{\ell} = \epsilon_h = 0.2$ 。更新过程通过参考路径惩罚对初始SFT策略 θ_0 进行正则化，

$$\mathcal{R}_{\text{ref}} = \exp(\Delta_i) - \Delta_i - 1, \quad \Delta_i = \log p_{\theta_0}(\tau_i \mid x, c_i) - \log p_{\theta}(\tau_i \mid x, c_i). \quad (7)$$

最终损失为 $\mathcal{L}_{\text{GRPO}} + \beta \mathcal{R}_{\text{ref}}$ ，其中 $\beta = 0.002$ 。 $\mathcal{R}_{\text{ref}}$ 是基于累积SDE路径对数概率计算的标准 k_3 KL估计量，充当香草参考策略惩罚。

4 实验

4.1 实验设置

我们在 WOD-E2E RFS 协议下，对 MINDVLA-U1 上的 DIAL 进行评估。第一阶段的 SFT 模型使用 Waymo 训练分割进行训练。偏好优化利用了 RFS 标注验证集

The reward is the rater feedback score:

$$R_i = \text{RFS}(\tau_i; \mathcal{P}_x), \quad (2)$$

where $\mathcal{P}_x$ denotes the rater trajectories and labels for scene x . Auxiliary geometric costs can be included, but the main implementation keeps them at zero and uses the preference reward described below. Advantages are normalized within the proposals of the same scene:

$$A_i = \frac{R_i - \frac{1}{K} \sum_{j=1}^K R_j}{\text{std}_{j=1}^K(R_j) + \epsilon}. \quad (3)$$

This is the step that makes intent diversity useful for RL: if all proposals express the same intent, the group has little preference contrast; if intents expose different intents, RFS supplies a meaningful within-scene ranking. Because the group always spans all eight intents rather than K noise variants of one, every intent class receives a preference signal in the same update.

3.3 Reward-Hacking-Aware RFS Reward

The reported evaluation metric remains the standard WOD-E2E RFS score. For training, however, using the evaluator unchanged creates two avoidable reward-hacking paths. Standard RFS scores only the 3s and 5s anchors, so intermediate waypoints can drift while the two anchors stay inside the acceptable boxes. It also applies a hard maximum over up to three rater trajectories, so a model can overfit to the easiest high-label rater geometry on the RL training split.

We therefore use a training-side RFS variant with the same rater preference inputs but different aggregation. First, the rater maximum is replaced by label-softmax aggregation. For rater label y_p , geometric decay $d_{p,a}$ at anchor a , and temperature τ , the per-anchor score is

$$\tilde{R}_a = \sum_p \frac{\exp(\tau y_p)}{\sum_q \exp(\tau y_q)} y_p d_{p,a}. \quad (4)$$

The weights depend only on fixed rater labels, not on model-controlled geometry, so the policy cannot manipulate the aggregation by moving toward one rater. We tune τ on the split-aware validation protocol and report the sensitivity in Section 4.6.2. Second, the training anchors are densified from $\{3, 5\}$ seconds to $\{1, 2, 3, 4, 5\}$ seconds. The additional anchors penalize discontinuous or implausible intermediate motion while preserving the canonical 3s and 5s scoring points used by evaluation.

3.4 Policy Update

Each sampled trajectory is replayed under the same intent condition used to generate it. Let $\log p_{\theta_{\text{old}}}(\tau_i \mid x, c_i)$ be the stored SDE path log-probability and $\log p_{\theta}(\tau_i \mid x, c_i)$ the replayed current log-probability. The clipped GRPO objective is

$$\rho_i = \exp(\log p_{\theta}(\tau_i \mid x, c_i) - \log p_{\theta_{\text{old}}}(\tau_i \mid x, c_i)), \quad (5)$$

$$\mathcal{L}_{\text{GRPO}} = -\frac{1}{K} \sum_{i=1}^K \min(\rho_i A_i, \text{clip}(\rho_i, 1 - \epsilon_{\ell}, 1 + \epsilon_h) A_i). \quad (6)$$

We use symmetric clipping with $\epsilon_{\ell} = \epsilon_h = 0.2$. The update is regularized against the starting SFT policy θ_0 using a reference path penalty,

$$\mathcal{R}_{\text{ref}} = \exp(\Delta_i) - \Delta_i - 1, \quad \Delta_i = \log p_{\theta_0}(\tau_i \mid x, c_i) - \log p_{\theta}(\tau_i \mid x, c_i). \quad (7)$$

The final loss is $\mathcal{L}_{\text{GRPO}} + \beta \mathcal{R}_{\text{ref}}$ with $\beta = 0.002$. $\mathcal{R}_{\text{ref}}$ is the standard k_3 KL estimator computed on the cumulative SDE path log-probability and acts as a vanilla reference-policy penalty.

4 Experiments

4.1 Experimental Setup

We evaluate DIAL on MINDVLA-U1 under the WOD-E2E RFS protocol. The first-stage SFT model is trained on the Waymo training split. Preference optimization uses the RFS-labeled validation

池。我们通过序列_id的确定性哈希将 438 个验证序列进行分割：其中 338 个序列用于强化学习训练，100 个序列留作评估，split_seed=43。以下所有 RL 对比均使用此分割。

主要实验从ckpt8k意图条件监督微调检查点开始，应用多意图CFG GRPO，使用 $C = 8$ 个意图和每个意图 $S = 2$ 个样本，为每个场景生成 $K = CS = 16$ 个候选方案。训练采用批次大小4、恒定学习率 5×10^{-7} 、噪声等级为0.5的CPS SDE采样、PPO裁剪0.2，以及在累积对数概率 k_3 KL估计器上使用参考路径系数 $\beta = 0.002$ 。训练侧奖励使用第3.3节中描述的奖励破解感知RFS变体；评估报告标准Waymo RFS。留出RFS是主要选择指标；全分割RFS和信任区域率(TR)描述所选检查点。

4.2 主要结果

基线训练协议。 对于表1，我们在统一的仅Waymo任务训练协议下重新训练并适配官方基线实现。监督微调阶段完全由Waymo训练分割通过我们的Waymo数据加载器构建，该加载器提供相机观测、自我运动上下文、导航意图和未来轨迹监督，并将其转换为每个模型所需的数据集格式。相同的Waymo监督微调源用于ReCogDrive [Li等人，2025b]、AutoVLA [Zhou等人，2025]、Curious-VLA [Chen等人，2026]和WAM-Flow [Xu等人，2025b]；在任务特定的监督微调中未使用非Waymo驾驶数据集。当应用强化学习时，每个策略从其Waymo-SFT检查点初始化，并在Waymo RFS子集上使用GRPO风格rollout和MindVLA/Waymo RFS奖励进行优化。因此，官方基线代码库和发布的权重仅作为架构实现或初始化使用，而我们实验中所有任务特定的监督微调数据和强化学习奖励数据均来自Waymo。

声明。 在这种受控的仅限Waymo的任务训练协议下，DIAL在表1的强化学习训练系统中产生了最强的留出改进。每个峰值均与同一强化学习阶段的步骤0评估进行比较，因此报告的增益纯粹归因于偏好优化，而非SFT训练进展。从MINDVLA-U1在留出RFS上达到7.696开始，DIAL达到了8.211（+0.515），全分割RFS从7.369上升至8.631。基线模型在同一留出划分上的改进较小：WAM-Flow提升 +0.087，Curious-VLA提升 +0.146，AutoVLA提升 +0.043，ReCogDrive提升 +0.315。DIAL还拥有最高的峰值留出分数和最大的TR增加（从54.7%到68.0%），而ReCogDrive是最强的基线峰值，达到7.714。这些比较与论文的核心机制一致：当意图条件采样暴露了RFS可以排名的操控级别替代方案（而不仅仅是一条记录路径的局部坐标变体）时，偏好优化最为有效。

表1：在仅Waymo监督微调和偏好优化后，在338/100留出划分（划分_种子=43）上的Waymo开放数据集端到端路线遵守分数。所有模型使用相同的特定任务Waymo监督微调来源。在每个块内，峰值行是具有最高留出RFS的强化学习检查点，起始行是同一强化学习阶段的步骤0评估； Δ_{RL} 是自起始以来的留出RFS变化；TR和全奖励函数分数位于峰值检查点。

模型	动作表示	阶段	留出RFS	Δ_{RL}	TR	全奖励函数分数
WAM-Flow	离散流	SFT初始化	5.547	—	24.0%	5.757
		强化学习峰值	5.634	+0.087	19.0%	6.111
Curious-VLA	动作标记	SFT初始化	5.808	—	30.0%	5.827
		强化学习峰值	5.954	+0.146	31.0%	7.157
AutoVLA	动作标记	SFT初始化	6.744	—	46.0%	6.809
		强化学习峰值	6.787	+0.043	47.0%	6.780
ReCogDrive	DiT扩散	SFT初始化	7.399	—	58.0%	7.244
		强化学习峰值	7.714	+0.315	65.4%	7.543
DIAL	连续流	SFT初始化	7.696	—	54.7%	7.369
		强化学习峰值	8.211	+0.515	68.0%	8.631

pool. We split the 438 validation sequences by a deterministic hash of sequence_id: 338 sequences are used for RL training and 100 sequences are held out for evaluation, with split_seed=43. All RL comparisons below use this split.

The main run starts from the ckpt8k intent-conditioned SFT checkpoint and applies multi-intent CFG GRPO with $C = 8$ intents and $S = 2$ samples per intent, giving $K = CS = 16$ proposals per scene. Training uses batch size 4, constant learning rate 5×10^{-7} , CPS SDE sampling with noise level 0.5, PPO clipping 0.2, and reference-path coefficient $\beta = 0.002$ on the cumulative-log-probability k_3 KL estimator. The training-side reward uses the reward-hacking-aware RFS variant described in Section 3.3; evaluation reports standard Waymo RFS. Held-out RFS is the main selection metric; full-split RFS and trust-region rate (TR) describe the selected checkpoint.

4.2 Main Results

Baseline training protocol. For Table 1, we retrain and adapt the official baseline implementations under a common Waymo-only task-training protocol. The supervised fine-tuning stage is constructed exclusively from the Waymo training split through our Waymo dataloader, which provides camera observations, ego-motion context, navigation intent, and future trajectory supervision, and converts them into the dataset format required by each model. The same Waymo SFT source is used for ReCogDrive [Li et al., 2025b], AutoVLA [Zhou et al., 2025], Curious-VLA [Chen et al., 2026], and WAM-Flow [Xu et al., 2025b]; no non-Waymo driving dataset is used for task-specific supervised fine-tuning. When RL is applied, each policy is initialized from its Waymo-SFT checkpoint and optimized on the Waymo RFS subset with GRPO-style rollouts and the MindVLA/Waymo RFS reward. Thus, official baseline codebases and released weights serve only as architectural implementations or initializations, while all task-specific SFT data and RL reward data in our experiments come from Waymo.

Claim. Under this controlled Waymo-only task-training protocol, DIAL produces the strongest held-out improvement among the RL-trained systems in Table 1. Each peak is compared with the step-0 evaluation from the same RL stage, so the reported gain isolates preference optimization rather than SFT training progress. Starting from MINDVLA-U1 at 7.696 held-out RFS, DIAL reaches 8.211 (+0.515), with full-split RFS rising from 7.369 to 8.631. The baselines improve less on the same held-out split: WAM-Flow gains +0.087, Curious-VLA +0.146, AutoVLA +0.043, and ReCogDrive +0.315. DIAL also has the highest peak held-out score and the largest TR increase (54.7% to 68.0%), while ReCogDrive is the strongest baseline peak at 7.714. These comparisons are consistent with the paper’s central mechanism: preference optimization is most effective when intent-conditioned sampling exposes maneuver-level alternatives that RFS can rank, rather than only local coordinate variants of a logged path.

Table 1: WOD-E2E RFS on the 338/100 held-out split (split_seed=43) after Waymo-only SFT and preference optimization. All models use the same task-specific Waymo SFT source. Within each block, the peak row is the RL checkpoint with the highest held-out RFS and the init row is the step-0 evaluation from the same RL stage; Δ_{RL} is the held-RFS change from init; TR and Full RFS are at the peak checkpoint.

Model	Action Representation	Stage	Held RFS	Δ_{RL}	TR	Full RFS
WAM-Flow	discrete flow	SFT init	5.547	—	24.0%	5.757
		RL peak	5.634	+0.087	19.0%	6.111
Curious-VLA	action token	SFT init	5.808	—	30.0%	5.827
		RL peak	5.954	+0.146	31.0%	7.157
AutoVLA	action token	SFT init	6.744	—	46.0%	6.809
		RL peak	6.787	+0.043	47.0%	6.780
ReCogDrive	DiT diffusion	SFT init	7.399	—	58.0%	7.244
		RL peak	7.714	+0.315	65.4%	7.543
DIAL	continuous flow	SFT init	7.696	—	54.7%	7.369
		RL peak	8.211	+0.515	68.0%	8.631

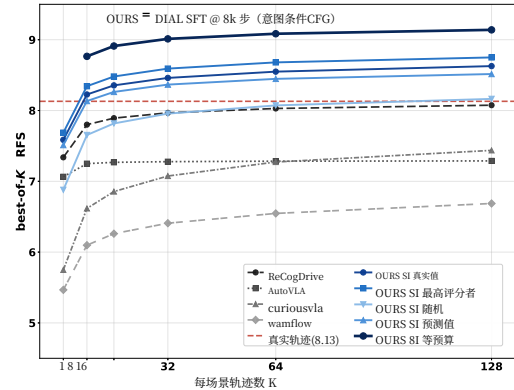


图3: 强化学习前提议上限。最佳- K RFS 与预算 K 。灰色虚线: 四个 SFT 基线均在 $K=128$ 处饱和于真实值 (8.13, 红色虚线) 之下。蓝色: 四种策略下的意图条件监督微调 (真实值、*top-rater*、预测值、随机), 均在 $K \approx 8$ 处超越真实值。深蓝色: 8意图等预算池化在 $K=128$ 处达到9.14。

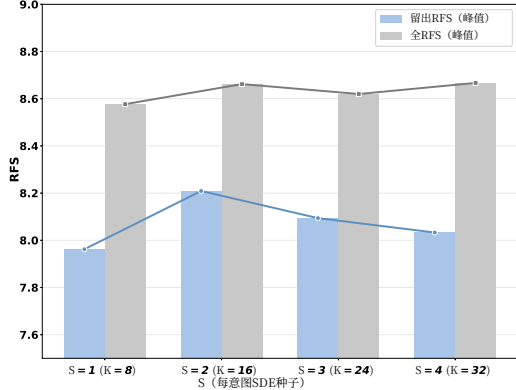


图4: 每个意图的样本数在 $C=8$ 。扫描 $S \in \{1, 2, 3, 4\}$ ($K \in \{8, 16, 24, 32\}$): 留出RFS (浅蓝色) 随 $S=2$ 呈非单调变化, 峰值高度最佳点达到 (8.21); 全分割RFS (浅灰色) 在 S 范围内几乎保持平坦。

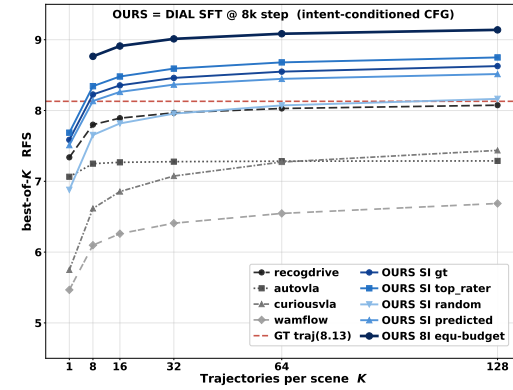


Figure 3: **Pre-RL proposal ceiling.** Best-of- K RFS vs. budget K . Gray dashed: four SFT base-lines all saturate below GT (8.13, red dashed) at $K=128$. Blue: intent-conditioned SFT under four strategies (*gt*, *top-rater*, *predicted*, *random*), all cross GT at $K \approx 8$. Navy: 8-intent equal-budget pooling reaches 9.14 at $K=128$.

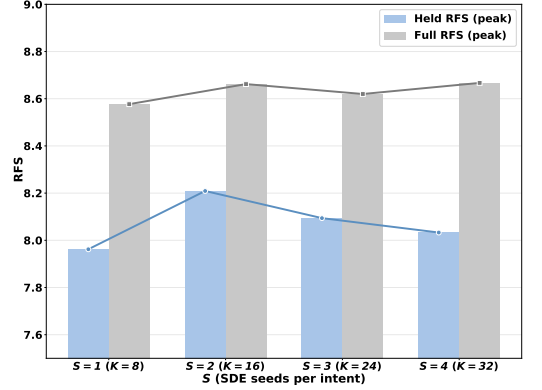


Figure 4: **Samples per intent at $C=8$.** Sweeping $S \in \{1, 2, 3, 4\}$ ($K \in \{8, 16, 24, 32\}$): held-out RFS (light blue) is non-monotone with $S=2$ as the peak-height sweet spot (8.21); full-split RFS (light gray) is nearly flat across S .

4.3 意图-CFG 提议上限

声明。 意图条件CFG扩展了提议支撑集, 超越了普通SFT采样, 在任意强化学习更新之前, 将最佳- N RFS上限提升至已记录的人类示范之上。

在同一批RFS标注场景上, 我们比较了三种提议支撑集: 记录轨迹、来自普通SFT策略 (四个竞争性的视觉语言动作基线: WAM-Flow、Curious-VLA、AutoVLA、ReCogDrive) 的最佳- N , 以及来自意图条件SFT策略在四种每个样本意图策略下的最佳- N 。图3绘制了随着 K 从1增长到128时的预期最佳- K RFS。记录轨迹得分为8.13 RFS (红色虚线参考)。所有四个基线视觉语言动作基线均在 $K = 128$ 处饱和于真实值线之下, 证实了从SFT策略进行的普通随机采样无法恢复人类评分员相较于记录轨迹更偏好的轨迹。采用任何单意图策略 (真实值、分类器预测、*top-rater*或随机) 的意图条件策略已在 $K \approx 8$ 处超越了真实值线, 因为意图条件化将提议引导至不同的语义盆地, 而非扰动单一的操纵模式。跨所有八个意图类别以每个意图等预算汇集提议 (最右侧曲线, $K = 8 \times n_{\text{per-intent}}$) 将上限扩展至 $K = 128$ 处的9.14, 相比最佳基线提升了 +1.07。

4.4 多意图 vs 单意图

声明。 在每个偏好组内覆盖完整的意图集是必要的; 任何单意图方案——无论条件意图如何选择——要么在DIAL达到峰值前崩塌, 要么陷入平台期。

我们保持所有其他设置与主要DIAL配方相同, 仅改变每个rollout的条件意图选择方式, 同时保持每场景预算为 $K = 16$ 。多意图GRPO (主要运行) 通过使用每个意图 $S = 2$ 个样本覆盖全部 8个意图类别, 为每个场景生成 $K = 16$ 个rollout。四个单意图基线使用 $C = 1$ 和 $S = 16$ 个噪声种子, 因此总预算保持为 $K = 16$ 。单意图变体则通过选择单一意图的方式来定义: 真实值 来自记录轨迹的几何意图; 预测值 来自部署意图分类器头; *top-rater* 来自评分最高的评分轨迹的几何信息 (一种标签泄露上界); 以及随机在每个批次中均匀随机抽取。

表2比较了相同总预算 $K = 16$ 下不同组构成之间的留出RFS峰值。DIAL达到了8.211的峰值 (+0.515 对比 SFT初始化)。四种单意图基线均未达到8.00, 并且最终都出现下降。随机作为最强的单意图基线, 峰值达到7.992, 但比DIAL低了0.22。预测达到了7.864的峰值, 但随后出现延迟性崩溃; 其训练集RFS

4.3 Intent-CFG Proposal Ceiling

Claim. Intent-conditioned CFG expands the proposal support beyond ordinary SFT sampling, raising the best-of- N RFS ceiling above the logged human demonstration before any RL update.

On the same RFS-labeled scenes, we compare three proposal supports: the logged trajectory, best-of- N from the ordinary SFT policy (four competitive VLA baselines: WAM-Flow, Curious-VLA, AutoVLA, ReCogDrive), and best-of- N from the intent-conditioned SFT policy under four per-sample intent strategies. Figure 3 plots the expected best-of- K RFS as K grows from 1 to 128. The logged trajectory scores 8.13 RFS (red dashed reference). All four baseline VLAs saturate *below* the GT line at $K = 128$, confirming that ordinary stochastic sampling from the SFT policy cannot recover trajectories the human rater would prefer over the log. The intent-conditioned policy with any single-intent strategy (*gt*, classifier-predicted, *top-rater*, or *random*) crosses the GT line already at $K \approx 8$, because intent conditioning guides proposals into distinct semantic basins rather than perturbing a single maneuver mode. Pooling proposals across all eight intent classes with equal per-intent budget (right-most curve, $K = 8 \times n_{\text{per-intent}}$) extends the ceiling to 9.14 at $K = 128$, an improvement of +1.07 over the best baseline.

4.4 Multi-Intent vs Single-Intent

Claim. Spanning the full intent set within each preference group is necessary; any single-intent recipe – regardless of how the conditioning intent is selected – either collapses or plateaus before DIAL’s peak.

We hold all other settings fixed at the main DIAL recipe and vary only how each rollout’s conditioning intent is chosen, while keeping the per-scene budget at $K = 16$. Multi-intent GRPO (the main run) draws $K = 16$ rollouts per scene by spanning all 8 intent classes with $S = 2$ per intent. The four single-intent baselines use $C = 1$ and $S = 16$ noise seeds, so the total budget remains $K = 16$. The single-intent variant is then defined by how the one intent is picked: *gt* from the geometric intent of the logged trajectory; *predicted* from the deployment intent classifier head; *top-rater* from the highest-rated rater trajectory’s geometry (a label-leakage upper bound); and *random* from a uniform draw at every batch.

Table 2 compares held-out RFS peaks across group compositions at the same total budget $K = 16$. DIAL reaches a peak of 8.211 (+0.515 vs SFT init). None of the four single-intent baselines reaches 8.00, and all ultimately decline. *Random*, the strongest single-intent baseline, peaks at 7.992 but falls 0.22 short of DIAL. *Predicted* reaches 7.864 with a delayed collapse; its training-split RFS

持续上升至8.156，而留出RFS却下降，这是奖励破解的典型标志。真实值和 $top-rater$ 的崩溃更为剧烈。在DIAL的峰值检查点，所有单意图变体均低于其0.67至1.98个RFS点。因此，覆盖完整意图集既是为了达到具有竞争力的峰值，也是为了在整个训练过程中维持偏好对比。所有变体的训练动态可视化结果见图5（附录B）。

表2：多意图组 vs 单意图基线，均共享主DIAL配方和每场景预算 $K = 16$ 。在各变体的保留峰值检查点上报告TR和全奖励函数分数。

组构成	C	S	K	保留峰值	TR	全奖励函数分数
真实值（单，几何形状）	1	16	16	7.783	65.0%	7.733
预测的（单一，分类器）	1	16	16	7.864	57.0%	8.331
$top-rater$ （单一，泄漏）	1	16	16	7.728	61.0%	7.545
随机（单一，均匀）	1	16	16	7.992	64.0%	8.035
DIAL（多意图，主变体）	8	2	16	8.211	68.0%	8.631

4.5 多样性保持分析

声明。 多意图CFG GRPO能够在偏好优化过程中防止提案多样性崩溃，从而保持阶段1建立的扩展采样分布。

We 通过在已训练检查点上进行推理计算多样性指标来支持该声明 s. No 需要进行额外的强化学习训练。

意图间提案距离(D1)和RFS离散度(D2)。 对于每个场景，我们在所有8个意图下进行解码（禁用CFG以获得纯意图条件样本），计算所有 $\binom{8}{2} = 28$ 轨迹对的成对平均位移误差作为D1，并以8条轨迹的RFS标准差作为D2。所有强化学习变体均在相同检查点（迭代4800）进行评估，以确保公平比较。表3显示，DIAL保持了最高的D2（0.75），这意味着在偏好优化之后，不同意图仍然能够产生质量有差异的轨迹。单意图变体表现出两种失败模式：随机模式在空间上坍缩（D1 = 2.40 m，最低），无论意图条件如何都会生成几乎相同的轨迹；而 $top-rater$ 模式则表现出空间离散（D1 = 7.08 m，最高），但质量区分度差（D2低）。多样性红利差距 = D3@16 − D3@1衡量了在16个意图条件提案中通过选择可以额外恢复多少RFS。DIAL保持了 +2.04 的差距（接近SFT初始化的 +2.23），而所有单意图变体与DIAL相比，差距损失在0.19到0.57之间。请注意，DIAL的D3@16（6.540）略低于SFT初始化（6.617）：强化学习将概率集中在得分更高的部署模式上，代价是最高16选多样性上限略有降低——这是预期的利用-多样性权衡，但不影响贪婪部署（留出RFS从7.696提升至8.211）。关键的是，差距排序与留出RFS排序完全一致：强化学习期间保留的多样性越多，最终的RFS就越高。

表3：迭代4800处公平跨变体比较的多样性指标。D1：每场景8条意图条件轨迹的成对平均位移误差。D2：相同8条轨迹的每场景RFS标准差。D3@1/D3@16：最佳1选和最佳16选预留RFS。Gap =D3@16 −D3@1：多样性红利。SFT初始化行使用任何强化学习之前的ckpt8k检查点。

采样	D1（米）	D2	D3@1	D3@16	Gap
SFT初始化（无强化学习）	6.43	0.52	4.387	6.617	+2.23
DIAL（多意图）	4.17	0.75	4.500	6.540	+2.04
单意图：随机	2.40	0.43	4.381	6.231	+1.85
单意图：预测	4.82	0.64	4.372	6.172	+1.80
单意图：最高评分	7.08	0.65	4.240	6.020	+1.78
单意图：真实意图	6.02	0.59	4.229	5.889	+1.66

continues rising to 8.156 while held-out RFS falls, the standard signature of reward hacking. Gt and $top-rater$ collapse more sharply. At DIAL’s peak checkpoint, all single-intent variants lie 0.67–1.98 RFS points below. Spanning the full intent set is therefore necessary both for reaching a competitive peak and for maintaining preference contrast throughout training. Training dynamics for all variants are visualized in Figure 5 (Appendix B).

Table 2: Multi-intent group vs single-intent baselines, all sharing the main DIAL recipe and per-scene budget $K = 16$. TR and Full RFS are reported at each variant’s held-peak checkpoint.

Group composition	C	S	K	Held peak	TR	Full RFS
gt (single, geometric)	1	16	16	7.783	65.0%	7.733
$predicted$ (single, classifier)	1	16	16	7.864	57.0%	8.331
$top-rater$ (single, leakage)	1	16	16	7.728	61.0%	7.545
$random$ (single, uniform)	1	16	16	7.992	64.0%	8.035
DIAL (multi-intent, main)	8	2	16	8.211	68.0%	8.631

4.5 Diversity Preservation Analysis

Claim. Multi-intent CFG GRPO prevents proposal diversity from collapsing during preference optimization, preserving the expanded sampling distribution established in Stage 1.

We support the claim with diversity metrics computed by inference on already-trained checkpoints. No additional RL training is required.

Inter-intent proposal distance (D1) and RFS spread (D2). For each scene we decode under all 8 intents (with CFG disabled to obtain pure intent-conditional samples), compute D1 as the mean pairwise ADE across the $\binom{8}{2} = 28$ trajectory pairs, and D2 as the RFS standard deviation across the 8 trajectories. All RL variants are evaluated at the same checkpoint (iter 4800) for a fair comparison. Table 3 shows that DIAL retains the highest D2 (0.75), meaning different intents still produce quality-differentiated trajectories after preference optimization. Single-intent variants show two failure modes: $random$ collapses spatially (D1 = 2.40 m, lowest), producing near-identical trajectories regardless of the intent conditioning, while $top-rater$ shows spatial scatter (D1 = 7.08 m, highest) but poor quality differentiation (low D2). The diversity dividend $gap = D3@16 - D3@1$ measures how much additional RFS is recoverable by selecting among 16 intent-conditioned proposals. DIAL preserves a gap of +2.04 (close to the SFT initialization’s +2.23), while all single-intent variants lose between 0.19 and 0.57 gap versus DIAL. Note that DIAL’s D3@16 (6.540) is slightly below the SFT initialization’s (6.617): RL concentrates probability toward better-scoring deployment modes at the cost of a small reduction in the best-of-16 diversity ceiling – the expected exploitation–diversity trade-off, which does not affect greedy deployment (held-out RFS improves from 7.696 to 8.211). Crucially, the gap ordering matches the held-out RFS ranking exactly: the more diversity is preserved during RL, the higher the final RFS.

Table 3: Diversity metrics at iter 4800 for a fair cross-variant comparison. D1: mean pairwise ADE across 8 intent-conditional trajectories per scene. D2: per-scene RFS std across the same 8 trajectories. D3@1/D3@16: best-of-1 and best-of-16 held-out RFS. Gap = D3@16 − D3@1: diversity dividend. The SFT init row uses the ckpt8k checkpoint before any RL.

Sampling	D1 (m)	D2	D3@1	D3@16	Gap
SFT init (no RL)	6.43	0.52	4.387	6.617	+2.23
DIAL (multi-intent)	4.17	0.75	4.500	6.540	+2.04
Single: random	2.40	0.43	4.381	6.231	+1.85
Single: predicted	4.82	0.64	4.372	6.172	+1.80
Single: top-rater	7.08	0.65	4.240	6.020	+1.78
Single: GT intent	6.02	0.59	4.229	5.889	+1.66

4.6 消融研究

4.6.1 意图内样本数

Claim. 在意图集 $C = 8$ 固定的情况下，每个意图的样本数 S 控制着一种权衡关系 f bet在意图内优势方差与奖励破解起始之间。

图4在固定 $C = 8$ 下扫描 $S \in \{1, 2, 3, 4\}$ ，得到总组大小 $K \in \{8, 16, 24, 32\}$ 。 $S = 1(K = 8)$ 为每个意图提供一个rollout，组成本最低；保留峰值为7.962，且运行在整个训练过程中保持稳定。 $S = 2(K = 16)$ ，主配置)达到最高峰值8.211。 $S = 3(K = 24)$ 达到峰值8.094， $S = 4(K = 32)$ 峰值在8.033，证实了更高的 S 会使峰值提前出现并略微降低峰值高度。在所有运行中， $S = 2$ 仍是峰值高度的最佳选择； $S = 1$ 仍是部署稳定性的工作点。

4.6.2 奖励塑形

标准 Waymo RFS 使用评分者轨迹的硬最大值，并且仅对 3秒和5秒锚点进行评分，这会在强化学习过程中产生可避免的奖励破解路径。我们用一个标签软最大值聚合（温度 τ ）取代了最大值，并将评分锚点加密到 $\{1, 2, 3, 4, 5\}$ 秒；表4对这些选择进行了消融研究。峰值高度排序在 τ 中非单调： $\tau = 0.3$ 给出了最高的保留峰值（8.211），而行 D（ $\tau = 1.0 +$ 密集锚点）在训练末期稳定性显著提升的情况下，实现了最强的完整划分数（8.728）。我们报告 $\tau = 0.3$ 作为实现峰值性能的主要配置；行 D 是适合部署的替代方案。

表4：在 338/100 留出划分（§ 4.1）上的训练侧奖励消融。所有变体共享多意图主配方（ $C = 8, S = 2, K = 16$ ）。TR 和 完整峰值 在保留峰值检查点处报告。

奖励变体	Aggr	锚点	τ	保留峰值	TR	完整峰值
A. 香草	max	稀疏 密集层	–	7.990	68.0%	8.328
B. 仅密集锚点	max		–	7.965	67.0%	8.430
C. 仅柔性最大值函数	柔性最大值函数	稀疏 密集层	1.0	8.147	72.0%	8.485
D. 柔性最大值函数 + 密集层（部署友好型）	柔性最大值函数		1.0	8.130	62.0%	8.728
柔性最大值函数 + 密集层 ($\tau=0.5$)	柔性最大值函数	密集层	0.5	8.097	62.0%	8.634
DIAL（主）	柔性最大值函数	密集层	0.3	8.211	68.0%	8.631
均值 ($\tau \rightarrow 0$) + 密集层	mean	密集层	–	7.834	60.0%	8.098

5 结论

我们识别出单演示监督微调模式坍缩是偏好强化学习在连续动作驾驶策略上的主要瓶颈：当采样轨迹聚集在一个操纵模式附近时，组相对奖励信号缺乏机动级对比，强化学习无法超越采样分布已有的内容提升。DIAL 通过两阶段训练流程解决了这个问题。阶段1 利用无分类器指导将扩散动作头与离散驾驶意图进行条件化，在任何强化学习更新之前将采样分布扩展为语义不同的机动盆地。阶段2 对每个场景的意图平衡的提议组运行多意图GRPO，通过偏好优化保持扩大的分布；任何单意图展开方案都会重新坍缩该组，并在训练结束时退化。这两个阶段共同将最佳 N RFS 上限提升至 $K=128$ 时的 9。14——超越了最强的先前规划器和人类驾驶演示——并将留出单轨迹 RFS 从 7。696 提高到 8。211，且没有任何单意图变体达到 8。00。

局限性。 当前实现的几个方面限制了结论的适用范围。首先，八种驾驶意图是通过手工编码规则从轨迹几何中派生出来的；它们涵盖了常见的机动类别，但可能无法涵盖不常见的❖❖尾行为，并且标记启发式可能错误分类模糊轨迹。其次，强化学习训练使用了包含438个序列的 RFS标记验证池，该池相对于完整的Waymo训练分割来说规模较小；奖励信号可能无法完全代表具有挑战性或罕见场景的分布。

4.6 Ablation Studies

4.6.1 Samples per Intent

Claim. Holding the intent set at $C = 8$ fixed, the number of samples per intent S controls a trade-off between within-intent advantage variance and reward-hacking onset.

Figure 4 sweeps $S \in \{1, 2, 3, 4\}$ at fixed $C = 8$, giving total group sizes $K \in \{8, 16, 24, 32\}$. $S = 1$ ($K = 8$) supplies one rollout per intent and gives the cheapest group; the held-out peak is 7.962 and the run remains stable throughout training. $S = 2$ ($K = 16$, the main configuration) reaches the highest peak 8.211. $S = 3$ ($K = 24$) reaches a peak of 8.094 and $S = 4$ ($K = 32$) peaks at 8.033, confirming that higher S shifts the peak earlier and slightly reduces peak height. Across all runs, $S = 2$ remains the peak-height sweet spot; $S = 1$ remains the deployment-stability operating point.

4.6.2 Reward Shaping

Standard Waymo RFS uses a hard maximum over rater trajectories and scores only the 3s and 5s anchors, creating avoidable reward-hacking paths during RL. We replace the maximum with a label-softmax aggregation (temperature τ) and densify the scored anchors to $\{1, 2, 3, 4, 5\}$ s; Table 4 ablates these choices. The peak-height ordering is non-monotone in τ : $\tau = 0.3$ gives the highest held-out peak (8.211), while row D ($\tau = 1.0 +$ dense anchors) achieves the strongest full-split score (8.728) with substantially better end-of-training stability. We report $\tau = 0.3$ as the main configuration for peak performance; row D is the deployment-friendly alternative.

Table 4: Training-side reward ablation on the 338/100 held-out split (§4.1). All variants share the multi-intent main recipe ($C = 8, S = 2, K = 16$). TR and *Full peak* are reported at the held-peak checkpoint.

Reward variant	Aggr	Anchor	τ	Held peak	TR	Full peak
A. Vanilla	max	sparse	–	7.990	68.0%	8.328
B. Dense anchors only	max	dense	–	7.965	67.0%	8.430
C. Softmax only	softmax	sparse	1.0	8.147	72.0%	8.485
D. Softmax + dense (deployment-friendly)	softmax	dense	1.0	8.130	62.0%	8.728
Softmax + dense ($\tau=0.5$)	softmax	dense	0.5	8.097	62.0%	8.634
DIAL (main)	softmax	dense	0.3	8.211	68.0%	8.631
Mean ($\tau \rightarrow 0$) + dense	mean	dense	–	7.834	60.0%	8.098

5 Conclusion

We identified single-demonstration SFT mode collapse as the primary bottleneck for preference RL on continuous-action driving policies: when sampled trajectories cluster around one maneuver mode, the group-relative reward signal carries no maneuver-level contrast, and RL cannot improve beyond what the sampling distribution already contains. DIAL addresses this with a two-stage training pipeline. Stage 1 conditions the diffusion action head on discrete driving intents with classifier-free guidance, expanding the sampling distribution into semantically distinct maneuver basins before any RL update. Stage 2 runs multi-intent GRPO over an intent-balanced proposal group per scene, preserving the expanded distribution through preference optimization; any single-intent rollout recipe re-collapses the group and degrades by training end. Together, the two stages lift the best-of- N RFS ceiling to 9.14 at $K=128$ – surpassing both the strongest prior planner and the human-driven demonstration – and improve held-out single-trajectory RFS from 7.696 to 8.211, with no single-intent variant reaching 8.00.

Limitations. Several aspects of the current implementation bound the scope of the conclusions. First, the eight driving intents are derived from trajectory geometry by hand-coded rules; they capture common maneuver categories but may not cover uncommon long-tail behaviors, and the labeling heuristic can misclassify ambiguous trajectories. Second, RL training uses the 438-sequence RFS-labeled validation pool, which is small relative to the full Waymo training split; the reward signal may not fully represent the distribution of challenging or rare scenarios.

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A 扩展相关工作

视觉-语言-动作策略。视觉-语言-动作模型将大型多模态表示适配于动作生成。RT-2研究了网络规模的视觉-语言预训练如何迁移至机器人控制 [Zitkovich 等, 2023]; OpenVLA 为机器人操作提供了开源VLA模型 [Kim 等, 2024]; 而 π_0 则采用基于流的动作模型进行通用机器人控制 [Black 等, 2024]。在自动驾驶领域, WOD-E2E定义了一个基于视觉的端到端基准, 包含具有挑战性的长尾场景和评分反馈评估 [Xu 等, 2025a], , 同时AutoVLA研究了一种具有自适应推理和强化微调能力的VLA驾驶模型 [Zhou 等,2025]。我们的工作并非主要追求更大的VLA架构, 而是提炼出一个提案支持瓶颈: 连续动作VLA只有在首先提出可供偏好信号排序的操纵替代方案时, 才能从偏好强化学习中获益。

意图与多模态轨迹提议。驾驶行为本质上是多模态的。Mul-tiPath 预测了锚点轨迹假设 [Chai 等人 2019], 上的一个分布, CoverNet 将行为预测视为对代表性轨迹集合 [Phan-Minh 等人 2020], 的分类, Tra-jectron++ 使用图结构生成模型进行动态可行的多智能体预测 [Salzmann 等人 2020], , 而 Motion Transformer 则将全局意图定位与局部轨迹精化相结合 [Shi 等人 2022]。这些方法表明, 未来运动更适合表示为一组结构化的备选方案, 而不是单一的回归目标。DIAL 将这一经验迁移到自车策略优化中: 意图标签不再是最终的预测目标, 而是一个控制变量, 迫使提议生成器在 RFS引导的强化学习之前暴露不同的操纵。

生成式连续动作策略。扩散模型与流模型是极具吸引力的动作头, 因为它们能够表示复杂的连续轨迹分布。扩散策略将视觉运动控制公式化为条件动作扩散 [Chi 等人 2025], , 而流匹配则通过沿着概率路径 [Lipman 等人2022]回归速度场来训练连续归一化流。近期的策略优化方法将这些生成器适配到奖励优化中, 包括用于扩散策略的 DPPO [Ren 等人 2024]、用于带有能量引导流策略的离线强化学习的 FlowQ [Alles 等人 2025]、用于流匹配策略在线微调的 ReinFlow [Zhang 等人, 2025c],以及流匹配策略梯度 [McAllister 等人 2025]。我们的设置是互补的: 我们保留了一个流风格的连续生成器, 但通过驾驶意图来结构化其采样群组, 使得奖励能够看到操纵级的变化, 而不仅仅是坐标级的噪声。

偏好优化与群组相对更新。偏好优化通常会在增加偏好输出可能性的同时, 使学习策略相对于参考策略进行正则化。PPO提供了裁剪策略梯度骨干 [Schulman等人, 2017], RLHF风格微调将学习到的人类偏好奖励应用于语言模型 [Ziegler等人, 2019], DPO通过直接优化成对偏好来消除显式的奖励模型强化学习 [Rafailov等人, 2023], 而GRPO则利用组相对优势来减少对学习值函数的需求 [Shao等人, 2024b]。WOD-E2E的RFS为轨迹而非文本提供了面向规划的偏好信号 [Xu等人, 2025a]。因此, DIAL保留了群组相对更新, 但将群组定义为来自同一场景的意图多样化的连续提议, 并在重放的连续生成路径上对更新进行正则化。

A Extended Related Work

Vision-language-action policies. Vision-language-action models adapt large multimodal representations to action generation. RT-2 studies how web-scale vision-language pretraining can transfer to robot control [Zitkovich et al., 2023]; OpenVLA provides an open-source VLA model for robotic manipulation [Kim et al., 2024]; and π_0 uses a flow-based action model for general robot control [Black et al., 2024]. In autonomous driving, WOD-E2E defines a vision-based end-to-end benchmark with challenging long-tail scenarios and rater-feedback evaluation [Xu et al., 2025a], while AutoVLA studies a VLA driving model with adaptive reasoning and reinforcement fine-tuning [Zhou et al., 2025]. Our work is not primarily a larger VLA architecture. It isolates a proposal-support bottleneck: a continuous-action VLA can only benefit from preference RL if it first proposes maneuver alternatives that the preference signal can rank.

Intent and multimodal trajectory proposals. Driving behavior is intrinsically multimodal. MultiPath predicts a distribution over anchor trajectory hypotheses [Chai et al., 2019], CoverNet casts behavior prediction as classification over a representative trajectory set [Phan-Minh et al., 2020], Tra-jectron++ uses a graph-structured generative model for dynamically feasible multi-agent forecasting [Salzmann et al., 2020], and Motion Transformer combines global intention localization with local trajectory refinement [Shi et al., 2022]. These methods show that future motion is better represented as a structured set of alternatives than as a single regression target. DIAL transfers that lesson to ego-policy optimization: the intent label is not the final prediction target, but a control variable that forces the proposal generator to expose distinct maneuvers before RFS-guided RL.

Generative continuous-action policies. Diffusion and flow models are attractive action heads because they can represent complex continuous trajectory distributions. Diffusion Policy formulates visuomotor control as conditional action diffusion [Chi et al., 2025], and flow matching trains continuous normalizing flows by regressing velocity fields along probability paths [Lipman et al., 2022]. Recent policy-optimization methods adapt these generators to reward optimization, including DPPO for diffusion policies [Ren et al., 2024], FlowQ for offline RL with energy-guided flow policies [Alles et al., 2025], ReinFlow for online fine-tuning of flow matching policies [Zhang et al., 2025c], and flow matching policy gradients [McAllister et al., 2025]. Our setting is complementary: we retain a flow-style continuous generator, but structure its sampling group by driving intent so that the reward sees maneuver-level variation rather than only coordinate-level noise.

Preference optimization and group-relative updates. Preference optimization commonly regularizes a learned policy against a reference policy while increasing the likelihood of preferred outputs. PPO supplies the clipped policy-gradient backbone [Schulman et al., 2017], RLHF-style fine-tuning applies learned human preference rewards to language models [Ziegler et al., 2019], DPO removes explicit reward-model RL by optimizing pairwise preferences directly [Rafailov et al., 2023], and GRPO uses group-relative advantages to reduce the need for a learned value function [Shao et al., 2024b]. WOD-E2E’s RFS provides a planning-facing preference signal for trajectories rather than text [Xu et al., 2025a]. DIAL therefore keeps the group-relative update, but defines the group as intent-diverse continuous proposals from the same scene and regularizes the update on the replayed continuous generation path.

B 训练动态

图5绘制了DIAL与来自第4.4节的四个单意图基线在整个强化学习训练过程中的留出RFS，它们共享相同的每场景预算 $K = 16$ 。

可以看到三种模式。首先，DIAL（多意图）的保留RFS峰值达到了8.211，之后仅适度下降，始终保持远高于所有单意图变体的水平。这种相对稳定性表明，意图多样化展开在每次迭代中都提供了有意义的组内偏好对比：该组始终包含语义上不同的操纵，因此奖励信号永远不会饱和。

其次，所有四种单意图变体的峰值都较低，且峰值后下降更为剧烈。随机达到了最高的单意图峰值（7.992），但在训练结束时远低于DIAL。真实值和最高评分者最早崩溃，这与以下观察结果一致：一旦策略学会在某个固定意图上获得高分，该意图的偏好对比便会迅速耗尽。

第三，预测 展现了典型的奖励攻击特征：训练集RFS在保留峰值后继续上升，而保留RFS却下降，这证实了优化与分类器预测对齐的单一意图会导致过度拟合强化学习训练划分，而非学习可迁移的偏好。这种训练集RFS与保留RFS之间的分歧在DIAL中不存在，因为涵盖所有意图为分布特定过拟合提供了结构屏障。

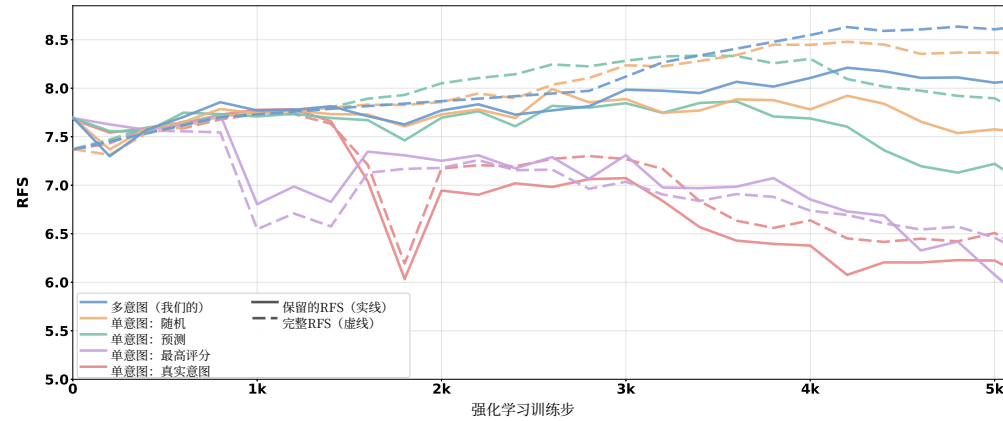


图5：整个强化学习训练过程中的留出RFS。DIAL（多意图， $C = 8$ ， $S = 2$ ）在所有变体中峰值最高且下降最少。单意图基线（ $C = 1$ ， $S = 16$ ）的峰值均较低且下降更剧烈。预测值（虚线）显示了奖励攻击特征：训练集RFS持续上升，而留出RFS在达到峰值后下降。所有变体共享相同的每场景预算 $K = 16$ ，并在338/100的留出划分上评估。

B Training Dynamics

Figure 5 plots held-out RFS throughout RL training for DIAL and the four single-intent baselines from Section 4.4, all sharing the same per-scene budget $K = 16$.

Three patterns are visible. First, DIAL (multi-intent) rises to its peak held-out RFS of 8.211 and subsequently declines only modestly, maintaining a substantially higher level than all single-intent variants throughout. The relative stability reflects that intent-diverse rollouts supply meaningful within-group preference contrast at every iteration: the group always spans semantically distinct maneuvers, so the reward signal never saturates.

Second, all four single-intent variants peak lower and exhibit sharper post-peak declines. *Random* reaches the highest single-intent peak (7.992) but falls well below DIAL by training end. *Gt* and *top-rater* collapse earliest, consistent with the observation that a fixed intent quickly exhausts preference contrast once the policy learns to score well on that intent.

Third, *predicted* displays the canonical reward-hacking signature: training-split RFS continues rising after the held-out peak while held-out RFS falls, confirming that optimizing a single intent aligned with the classifier’s prediction overfits to the RL training split rather than learning transferable preferences. This divergence between training-split and held-out RFS is absent in DIAL, where spanning all intents provides a structural barrier against distribution-specific overfitting.

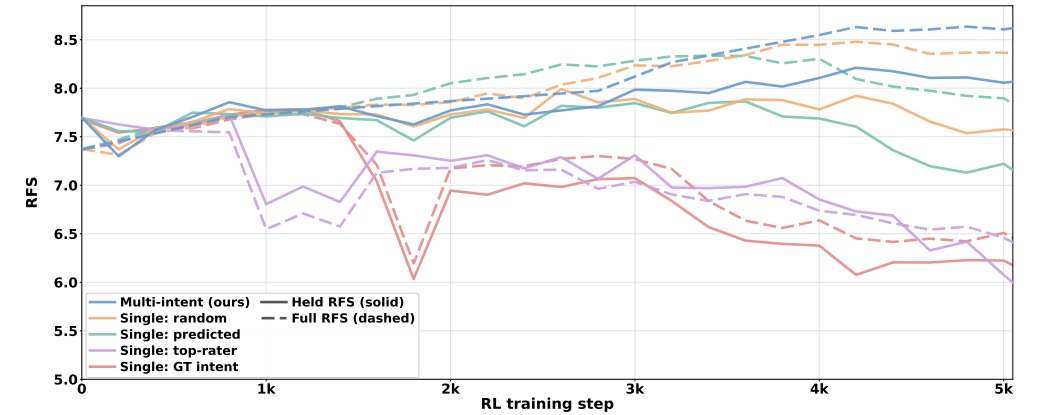


Figure 5: **Held-out RFS throughout RL training.** DIAL (multi-intent, $C = 8$, $S = 2$) peaks highest and declines least among all variants. Single-intent baselines ($C = 1$, $S = 16$) all peak lower and collapse more sharply. *Predicted* (dashed) shows the reward-hacking signature: training-split RFS continues rising while held-out RFS falls after the peak. All variants share the same per-scene budget $K = 16$ and are evaluated on the 338/100 held-out split.